

Key Ehrhart Data for $\lambda = (3, 2, 1, 0)$

We record the key Ehrhart data for $\lambda = (3, 2, 1, 0)$ and all $\sigma \in S_4$. Since trailing zero parts are suppressed in the database, the shape is stored as $\lambda = (3, 2, 1)$.

σ	dim	polynomial	h^*	ms
(1, 2, 3, 4)	0	1	[1]	0
(1, 2, 4, 3)	1	$k + 1$	[1, 0]	0
(1, 3, 2, 4)	1	$k + 1$	[1, 0]	0
(1, 3, 4, 2)	2	$(3/2)k^2 + (5/2)k + 1$	[1, 2, 0]	0
(1, 4, 2, 3)	2	$(3/2)k^2 + (5/2)k + 1$	[1, 2, 0]	0
(1, 4, 3, 2)	3	$k^3 + 3k^2 + 3k + 1$	[1, 4, 1, 0]	0
(2, 1, 3, 4)	1	$k + 1$	[1, 0]	0
(2, 1, 4, 3)	2	$k^2 + 2k + 1$	[1, 1, 0]	0
(2, 3, 1, 4)	2	$(3/2)k^2 + (5/2)k + 1$	[1, 2, 0]	0
(2, 3, 4, 1)	3	$(8/3)k^3 + 6k^2 + (13/3)k + 1$	[1, 10, 5, 0]	1
(2, 4, 1, 3)	3	$(7/3)k^3 + (11/2)k^2 + (25/6)k + 1$	[1, 9, 4, 0]	1
(2, 4, 3, 1)	4	$2k^4 + 7k^3 + 9k^2 + 5k + 1$	[1, 19, 25, 3, 0]	7
(3, 1, 2, 4)	2	$(3/2)k^2 + (5/2)k + 1$	[1, 2, 0]	0
(3, 1, 4, 2)	3	$2k^3 + 5k^2 + 4k + 1$	[1, 8, 3, 0]	1
(3, 2, 1, 4)	3	$k^3 + 3k^2 + 3k + 1$	[1, 4, 1, 0]	0
(3, 2, 4, 1)	4	$(23/12)k^4 + (20/3)k^3 + (103/12)k^2 + (29/6)k + 1$	[1, 18, 24, 3, 0]	6
(3, 4, 1, 2)	4	$(13/4)k^4 + (29/3)k^3 + (43/4)k^2 + (16/3)k + 1$	[1, 25, 44, 8, 0]	10
(3, 4, 2, 1)	5	$(11/6)k^5 + (49/6)k^4 + (29/2)k^3 + (77/6)k^2 + (17/3)k + 1$	[1, 38, 120, 58, 3, 0]	41
(4, 1, 2, 3)	3	$(8/3)k^3 + 6k^2 + (13/3)k + 1$	[1, 10, 5, 0]	1
(4, 1, 3, 2)	4	$(23/12)k^4 + (20/3)k^3 + (103/12)k^2 + (29/6)k + 1$	[1, 18, 24, 3, 0]	7
(4, 2, 1, 3)	4	$2k^4 + 7k^3 + 9k^2 + 5k + 1$	[1, 19, 25, 3, 0]	6
(4, 2, 3, 1)	5	$(7/3)k^5 + (39/4)k^4 + (49/3)k^3 + (55/4)k^2 + (35/6)k + 1$	[1, 43, 150, 81, 5, 0]	62
(4, 3, 1, 2)	5	$(11/6)k^5 + (49/6)k^4 + (29/2)k^3 + (77/6)k^2 + (17/3)k + 1$	[1, 38, 120, 58, 3, 0]	47
(4, 3, 2, 1)	6	$k^6 + 6k^5 + 15k^4 + 20k^3 + 15k^2 + 6k + 1$	[1, 57, 302, 302, 57, 1, 0]	299

For this example, the dimension agrees with the inversion number $\text{inv}(\sigma)$. It is also useful to record the sum of the h^* -entries.

σ	$\text{inv}(\sigma)$	$\sum_i h_i^*$
(1, 2, 3, 4)	0	1
(1, 2, 4, 3)	1	1
(1, 3, 2, 4)	1	1
(1, 3, 4, 2)	2	3
(1, 4, 2, 3)	2	3
(1, 4, 3, 2)	3	6
(2, 1, 3, 4)	1	1
(2, 1, 4, 3)	2	2
(2, 3, 1, 4)	2	3

σ	$\text{inv}(\sigma)$	$\sum_i h_i^*$
(2, 3, 4, 1)	3	16
(2, 4, 1, 3)	3	14
(2, 4, 3, 1)	4	48
(3, 1, 2, 4)	2	3
(3, 1, 4, 2)	3	12
(3, 2, 1, 4)	3	6
(3, 2, 4, 1)	4	46
(3, 4, 1, 2)	4	78
(3, 4, 2, 1)	5	220
(4, 1, 2, 3)	3	16
(4, 1, 3, 2)	4	46
(4, 2, 1, 3)	4	48
(4, 2, 3, 1)	5	280
(4, 3, 1, 2)	5	220
(4, 3, 2, 1)	6	720

If we trim the trailing zero from each h^* -vector and compare with the descent polynomials of principal weak and Bruhat ideals in S_4 , then we do find some coincidences.

Matches for the same permutation σ .

- (1, 2, 3, 4): [1], agreeing with both weak and Bruhat.
- (1, 3, 4, 2), (1, 4, 2, 3), (2, 3, 1, 4), (3, 1, 2, 4): [1, 2], agreeing with the weak descent polynomial.
- (1, 4, 3, 2) and (3, 2, 1, 4): [1, 4, 1], agreeing with both weak and Bruhat.

Cross-permutation coincidences.

- (2, 1, 4, 3) has trimmed $h^* = [1, 1]$, which equals the weak/Bruhat descent polynomial for (1, 2, 4, 3), (1, 3, 2, 4), and (2, 1, 3, 4).
- (2, 4, 1, 3) has trimmed $h^* = [1, 9, 4]$, which equals the Bruhat descent polynomial for (3, 4, 1, 2).
- (3, 1, 4, 2) has trimmed $h^* = [1, 8, 3]$, which equals the Bruhat descent polynomial for (2, 4, 3, 1) and (4, 2, 1, 3).

No trimmed h^* -vector of degree at least 4 agrees with a weak or Bruhat descent polynomial in S_4 , since those descent polynomials have degree at most 3.